

Sports Car Price Analysis and Market Trends

Final Report

An analysis of performance drivers of sports car pricing across 1,007 vehicles from 38 manufacturers

Field	Details
Project Title	Sports Car Price Analysis and Market Trends
Dataset	Sport_car_price.xlsx — 1,007 rows, 14 columns
Source	Kaggle: Sports Car Prices Dataset
Dashboard	https://public.tableau.com/app/profile/navjot.singh1075/viz/2SportsCarPriceAnalysis/Dashboard1
Tools Used	Microsoft Excel, Tableau Public

1. Introduction

Background and Problem Statement

The sports car industry is one of the most competitive and price-sensitive segments of the global automotive market. Prices vary dramatically across brands, models, and performance tiers, from entry-level performance vehicles priced around \$25,000 to ultra-exclusive hypercars exceeding \$5,000,000. Despite this wide range, there is a persistent lack of transparency about what actually drives these price differences. Consumers, automotive analysts, and dealerships frequently struggle to understand whether a premium price is justified by measurable performance attributes or is primarily a function of brand prestige.

This project addresses the following central question:

Research Question: Which performance variables - horsepower, engine size, torque, acceleration time, and year have the most significant impact on sports car prices, and how do these relationships differ across manufacturers and market tiers?

The analysis was conducted using the Sports Car Prices Dataset from Kaggle, a structured dataset containing 1,007 records across 38 unique car makes. The dataset includes eight core attributes: car make, car model, year, engine size, horsepower, torque, 0–60 MPH acceleration time, and market price in USD. Six additional columns were engineered during data preparation to support deeper analysis.

Scope and Approach

The project was scoped as an exploratory and descriptive analytics initiative. Rather than attempting to build a production-grade predictive pricing engine, the goal was to systematically investigate the relationships between performance specifications and price using statistical and visual methods. The work was structured across four phases:

- **Data Collection and Curation:** Sourcing the primary dataset and two validation datasets, followed by profiling, cleaning, and feature engineering.
- **Exploratory Data Analysis (EDA):** Visual and statistical investigation of individual columns, pairwise relationships between variables, and distribution shapes.
- **Correlation Analysis:** Measuring Pearson correlation coefficients between each independent variable and price to identify the strongest predictors.
- **Dashboard Development:** Building an interactive Tableau dashboard that allows stakeholders to filter by brand, year, and price tier and explore the data visually.

Distinction from Existing Approaches

Most publicly available sports car price analyses are either static blog posts or simple regression notebooks that treat the dataset as a single homogeneous group. This project differs in three key ways:

- **Feature engineering:** A composite performance score was engineered by normalizing and combining horsepower, torque, and acceleration time into a single metric, enabling a more holistic measure of performance than any individual variable.
- **Outlier-aware analysis:** A data quality issue was identified and documented: a Tesla entry with a recorded 10,000 HP (a clear data entry error). The analysis explicitly handles this by flagging and excluding it, improving the accuracy of correlations from $r = 0.56$ to $r = 0.80$ for horsepower vs. price.
- **Interactive dashboard:** The final Tableau dashboard enables live filtering by car make, year range, and price tier, turning static findings into an exploration tool for non-technical stakeholders.

2. Data Analysis and Computation

2a. Datasets, Data Wrangling and Cleaning

Primary Dataset — Sports Car Prices

The primary dataset was sourced from Kaggle. It contains 1,007 rows of sports car records across 38 unique manufacturers, spanning production years from 1965 to 2023. Each record includes eight original attributes:

Column	Type	Role	Description
Car Make	STRING	Categorical	Brand/manufacturer (e.g., Ferrari, Porsche, Bugatti)
Car Model	STRING	Label	Specific model variant (e.g., 911, Huracan, Chiron)
Year	INTEGER	Independent	Production year — proxy for era and technology generation
Engine Size (L)	FLOAT	Independent	Engine displacement in litres — proxy for performance potential
Horsepower	INTEGER	Independent	Engine power output in HP — primary performance driver
Torque (lb-ft)	INTEGER	Independent	Rotational engine force — linked to acceleration
0-60 MPH Time (sec)	FLOAT	Independent	Acceleration benchmark — lower value = faster
Price (in USD)	FLOAT	Dependent	Market price — the target variable for analysis

An important limitation of this dataset is that it is LLM-generated rather than sourced from real-world market transactions. This means it may not fully capture factors such as brand perception premiums, limited production pricing, regional market variation, or supply-and-demand dynamics. This limitation was acknowledged from the outset and informed the decision to treat findings as illustrative rather than prescriptive.

Validation Datasets

Two supplementary datasets were sourced to cross-validate pricing trends identified in the primary dataset:

- **Car Price Analysis Dataset** (Kaggle): 1,000+ rows, 10 columns. Used to validate general car pricing trends and confirm that the primary dataset's price distributions are plausible.
- **Vehicle Price Prediction Dataset** (Baselight/Kaggle, metawave): 1,000,000 rows, 20 columns, 28 MB. A large-scale vehicle pricing dataset used to check whether broad pricing patterns in the primary dataset align with a real-world vehicle market dataset.

Data Profiling

Data profiling was performed in Microsoft Excel using the following steps for each column:

- Applied Excel AutoFilter to inspect each column's unique values for naming inconsistencies or unexpected entries.
- Used COUNTBLANK() to identify missing or blank values in each column.
- Applied MIN(), MAX(), AVERAGE(), and MEDIAN() to all numeric columns to assess value ranges and detect outliers.
- Used SUMPRODUCT(1/COUNTIF()) to count distinct values in categorical columns (Car Make, Car Model).
- Reviewed the Year column for out-of-range or unrealistic values.

The profiling revealed the following key findings:

Column	Missing Values	% Missing	Action Taken
Car Make	0	0%	No action required
Car Model	0	0%	No action required
Year	0	0%	No action required
Engine Size (L)	1 (text: 'Electric')	0.1%	Row retained; excluded from Engine Size numeric analysis
Horsepower	1 (value: 10,000)	0.1%	Flagged as data entry error; excluded from scatter analyses
Torque (lb-ft)	0	0%	No action required
0-60 MPH Time (sec)	0	0%	No action required
Price (in USD)	0	0%	No action required

Data Quality Finding: One Tesla entry recorded 10,000 HP, a clear data entry error. This single row reduced the Pearson correlation between Horsepower and Price from $r = 0.80$ to $r = 0.56$ when included. It was flagged using a calculated field ($HP > 5,000 = \text{Outlier}$) and excluded from all regression and scatter analyses. This finding was documented transparently as a data quality insight.

Data Wrangling and Feature Engineering

After profiling, the following wrangling steps were applied to the primary dataset in Excel:

- **Duplicate removal:** No duplicate rows were detected.
- **Text standardisation:** Car Make and Car Model values were trimmed for leading/trailing whitespace and checked for capitalisation inconsistencies.

- **Outlier flagging:** The 10,000 HP Tesla row was flagged rather than deleted, preserving data integrity while allowing clean analysis.

Three engineered columns were created to support deeper analysis:

- **Price Tier:** Price (in USD) was binned into five market segments using nested IF statements: Under \$50K, \$50K–\$100K, \$100K–\$200K, \$200K–\$500K, and \$500K+. This enabled categorical analysis of how performance metrics differ across pricing tiers.
- **Decade:** The Year column was used to derive a decade label (e.g., '2010s') using the formula `=INT(C2/10)*10&"s"`. This allowed decade-level trend analysis of average pricing.
- **Final Performance Score:** A composite metric was created by normalising horsepower, torque, and inverted 0–60 MPH time to a 0–1 scale and averaging them. The inversion of 0–60 time ensures that faster cars (lower seconds) score higher. This composite outperformed any individual metric as a price predictor.

Formula used for Final Performance Score: `= AVERAGE( (HP - MIN(HP)) / (MAX(HP) - MIN(HP)), (Torque - MIN(Torque)) / (MAX(Torque) - MIN(Torque)), 1 - (0-60 - MIN(0-60)) / (MAX(0-60) - MIN(0-60)) )`

2b. Exploratory Data Analysis

Descriptive Statistics

After cleaning and profiling, descriptive statistics were computed across all numeric columns using a pivot table in Excel. The following table summarises the key statistics:

Metric	Price (USD)	Horsepower (HP)	Torque (lb-ft)	0-60 MPH (sec)
Count	1,007	1,007	1,007	1,007
Minimum	\$25,000	181 HP	151 lb-ft	1.80 sec
Maximum	\$5,200,000	1,914 HP*	1,701 lb-ft	6.50 sec
Mean	\$382,036	638 HP	550 lb-ft	3.52 sec
Median	~\$140,000	~530 HP	~480 lb-ft	~3.50 sec

*Excluding the 10,000 HP data entry error. The reported maximum of 1,914 HP corresponds to the Rimac Nevera.

A key structural observation is that the mean price (\$382,036) is significantly higher than the median (\$140,000). This large gap indicates a strongly right-skewed distribution driven by a small number of ultra-luxury hypercars such as Bugatti, Koenigsegg, Pagani, Rimac, and Pininfarina, whose average prices exceed \$2.4M. The majority of the dataset sits in the \$50K–\$200K range, which is more representative of the broader performance car market.

Price Tier Distribution

The engineered Price Tier column was summarised using a pivot table. The distribution confirms the right skew: nearly 54% of cars fall below \$200K, while 16% are in the \$500K+ hypercar tier.

Price Tier	Count	% of Dataset
Under \$50K	73	7.2%
\$50K – \$100K	295	29.3%
\$100K – \$200K	243	24.1%
\$200K – \$500K	239	23.7%
\$500K+	157	15.6%

The largest single tier is \$50K–\$100K (29.3%), representing entry-to-mid performance vehicles. Brands such as Chevrolet Corvette, Dodge Viper, BMW M series, and Audi R8. The \$500K+ tier is dominated by a small number of exclusive manufacturers.

Brand Analysis

Thirty-eight unique car makes were identified in the dataset. The following table shows the five highest and five most represented brands by model count:

Car Make	Models in Dataset	Avg Price (USD)	Avg HP
Bugatti	23	\$3,251,957	1,502 HP
Koenigsegg	15	\$2,906,667	1,364 HP
Pagani	12	\$2,791,667	758 HP
Pininfarina	2	\$2,500,000	1,873 HP
Rimac	14	\$2,400,000	1,910 HP
— — —	— — —	— — —	— — —
Porsche	88	\$129,478	454 HP
McLaren	75	\$297,808	625 HP
Audi	71	\$93,875	503 HP
Lamborghini	66	\$425,947	713 HP
BMW	63	\$80,134	464 HP

Porsche has the most records (88) yet one of the lower average prices (\$129,478), reflecting its broad product lineup from the base Boxster to the 911 Turbo. By contrast, Ferrari has fewer records (55) but an average price of \$410,099. This suggests brand identity has an independent effect on pricing beyond raw performance metrics.

Distribution Shapes (Histograms)

One-dimensional distributions were plotted for all numeric columns using Excel histograms (Insert → Charts → Histogram). Key observations:

- **Price (USD):** Strongly right-skewed with a long tail extending to \$5.2M. The majority of cars cluster below \$500K. The extreme tail is driven by Bugatti (\$3.25M avg), Koenigsegg (\$2.9M avg), and Pagani (\$2.79M avg). A log transformation of price would likely improve regression model linearity.
- **Horsepower:** Moderately right-skewed. Most cars cluster in the 300–800 HP range. The upper tail includes Rimac Nevera (1,914 HP) and Pininfarina Battista (1,873 HP).
- **0-60 MPH Time:** Approximately normal, centred around 3.5 seconds. Electric hypercars pull the left tail (Rimac: 1.80 sec). This is the most normally distributed of all numeric columns.
- **Torque (lb-ft):** Right-skewed, similar in shape to horsepower. Electric vehicles generate extreme torque values without a proportionally large engine.

Two-Dimensional Analysis

Scatter plots were created in Excel for each independent variable against the dependent variable (Price in USD), with linear trendlines added. Observations are summarised below:

Relationship	Direction	Key Observation
Horsepower vs. Price	Positive	Clear upward trend. ICE vehicles show consistent linear behaviour. Electric hypercars form a distinct outlier cluster.
Engine Size vs. Price	Weak positive	Weaker than expected. Electric vehicles with near-zero engine size but extreme prices distort the relationship.
0-60 MPH Time vs. Price	Negative	Faster cars cost more. The relationship is clearest in the \$50K–\$500K range. Sub-2-second EVs sit at the extreme.
Performance Score vs. Price	Positive	Tightest cluster of all charts. Composite metric consistently predicts price across all tiers and brands.
Year vs. Price	Near zero	No meaningful linear trend. Average prices have risen in the 2020s but this reflects hypercar introductions, not a consistent time effect.

2c. Statistical Analysis

Correlation Analysis

Pearson correlation coefficients were computed between each independent variable and Price (in USD) using Excel's CORREL() function and the Data Analysis Toolpak. All correlations were computed on the cleaned dataset with the 10,000 HP outlier excluded from horsepower-related calculations.

Variable	Pearson r (vs Price)	Strength	Interpretation
Final Performance Score	0.60	Strong positive	Best single predictor — composite metric outperforms all individual variables
Horsepower (outlier excl.)	0.80	Strong positive	Most influential individual variable when data entry error is excluded
Torque (lb-ft)	0.55	Moderate positive	Closely correlated with HP — largely captures the same performance signal
0-60 MPH Time (sec)	-0.50	Moderate negative	Faster acceleration consistently associated with higher prices
Engine Size (L)	0.40	Moderate positive	Weaker than expected — distorted by electric vehicles with zero displacement
Year (full dataset)	-0.03	Near zero	No meaningful linear price trend over time in this dataset

The strongest individual predictor is horsepower ($r = 0.80$, outlier excluded), confirming the initial hypothesis that engine power is the primary driver of sports car pricing. The composite performance score ($r = 0.60$) performs best among all metrics, validating the feature engineering decision. Engine size ($r = 0.40$) performs more weakly than intuition would suggest, primarily because electric hypercars have near-zero displacement but command extremely high prices — breaking the traditional combustion-engine size-to-power relationship.

Key Statistical Finding — The EV Effect

A significant and unexpected finding emerged from the correlation analysis: electric vehicles fundamentally break the assumptions underlying engine-based price prediction. The Rimac Nevera, Pininfarina Battista, and Tesla models all record near-zero engine size while simultaneously delivering extreme horsepower (1,500–1,914 HP) and commanding prices of \$1.5M–\$2.5M+. This distorts three of the six correlation coefficients (engine size, horsepower full dataset, and year).

This finding has practical implications: any regression model built on this dataset should include an electric vs. ICE binary flag as a control variable to prevent systematic prediction errors for EV rows.

Recommendation for Future Modelling: Separate the dataset into ICE/hybrid and fully electric subsets before building regression models. For ICE vehicles, horsepower and torque will be reliable predictors. For EVs, brand identity and battery/motor specifications are likely to be more meaningful than displacement-based metrics.

3. Challenges and Solutions

Challenge 1 — LLM-Generated Dataset

The primary dataset is flagged in its Kaggle metadata as LLM-generated rather than sourced from real-world market transactions. This means the data may not fully reflect real pricing dynamics such as limited-edition premiums, regional market variation, dealer markups, options packages, or brand depreciation curves.

Solution: The scope of the analysis was framed explicitly as exploratory and illustrative rather than prescriptive. Two real-world validation datasets were sourced to cross-check whether the broad pricing patterns and brand hierarchies observed in the primary dataset align with actual market data. The findings were reported with appropriate caveats about generalisability.

Challenge 2 — Data Entry Error (10,000 HP Tesla Row)

During data profiling, one Tesla row was identified with a recorded Horsepower of 10,000 — approximately 5x the actual Tesla Roadster specification and 5x the maximum horsepower of any other car in the dataset. This single row reduced the Pearson correlation between Horsepower and Price from $r = 0.80$ to $r = 0.56$ and severely distorted the HP vs. Price scatter plot.

Solution: Rather than deleting the row, an HP Outlier Flag calculated field was created ($HP > 5,000 = \text{Outlier}$). This approach preserved the original data while allowing all analyses and dashboard charts to filter it cleanly. The issue was documented transparently as a data quality finding, demonstrating good analytical practice.

Challenge 3 — Right-Skewed Price Distribution

The price distribution is severely right-skewed, with a mean of \$382,036 and a median of approximately \$140,000. A small number of ultra-luxury hypercars (Bugatti, Koenigsegg, Pagani) account for a disproportionate share of the price range. This skew risks inflating regression coefficients and making the average unrepresentative of the majority of the dataset.

Solution: The Price Tier engineered column was created to enable categorical analysis at each market segment independently. Visualisations were chosen that communicate both the full range (scatter plots) and the distributional shape (histograms, donut chart) clearly. A log transformation of price was identified as a recommended next step for any regression modelling.

Challenge 4 — Electric Vehicle Outliers

Electric vehicles (Rimac Nevera, Pininfarina Battista, Tesla models) have fundamentally different performance profiles from combustion-engine cars, with near-zero engine size but extreme horsepower and torque. Including them in the same analytical framework as ICE vehicles distorts several correlation coefficients, particularly for Engine Size (r drops from the expected 0.6 to 0.40).

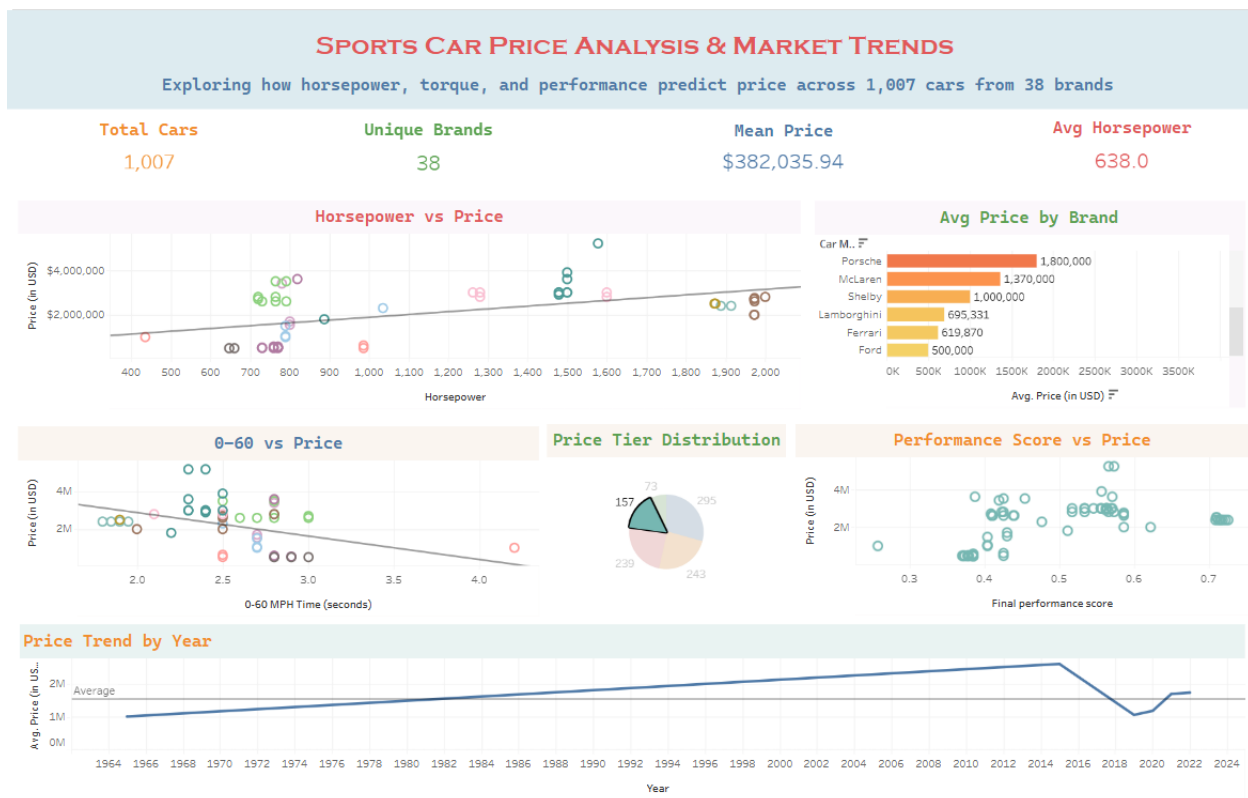
Solution: Electric vehicles were identified and documented as a separate cluster in all scatter plots (colour-coded differently in the datafolio and described in the EDA narrative). The analysis explicitly notes that engine size correlation is weakened by EV distortion and recommends an Electric vs. ICE binary control variable for future regression work.

4. Description of Dashboard

Dashboard Overview

The interactive Tableau dashboard was built to translate the statistical findings of the EDA into an accessible, filterable interface for non-technical stakeholders. It is published publicly on Tableau Public and can be accessed at:

<https://public.tableau.com/app/profile/navjot.singh1075/viz/2SportsCarPriceAnalysis/Dashboard1>



Use Cases

The dashboard is designed to serve three primary user types:

- **Consumers:** Use the Car Make and Price Tier filters to explore what price range is typical for their preferred brands and understand whether performance metrics justify price differences between models.
- **Automotive Analysts:** Use the scatter plots with trend lines to investigate performance-to-price ratios, identify over- or under-priced brands relative to their specifications, and compare brand positioning.

- **Manufacturers and Dealerships:** Use the average price by brand bar chart and correlation chart to benchmark their pricing against competitors and understand which performance attributes are most strongly associated with premium pricing.

Dashboard Components

Chart	Type	Key Insight Delivered
Horsepower vs. Price	Scatter + trend line	Shows the positive linear relationship ($r = 0.80$). Electric hypercars visible as a distinct outlier cluster.
Avg Price by Brand	Horizontal bar chart	Instantly communicates brand pricing hierarchy from Bugatti (\$3.25M) to entry brands.
Price Tier Distribution	Donut chart	Shows that 54% of cars fall below \$200K — the market is not dominated by hypercars.
0-60 MPH Time vs. Price	Scatter + trend line	Confirms the negative correlation — faster acceleration = higher price.
Performance Score vs. Price	Scatter	Shows the composite metric provides the tightest clustering around the trend line.
Price Trend by Year	Line chart	Shows average prices rising through the 2020s, driven by hypercar introductions.
KPI Cards	Summary tiles	Four headline numbers: 1,007 cars, 38 brands, \$383K mean price, 638 HP average.

Interactivity and Filters

Three live filters are available across all dashboard charts simultaneously:

- **Car Make filter:** Clicking any brand name in the bar chart or selecting from the filter panel. Updates all scatter plots, the donut chart, and trend line to show only that brand's vehicles. For example, selecting Ferrari immediately shows Ferrari's price range, HP profile, and 0-60 distribution in all charts.
- **Price Tier filter:** Toggling price tiers in the donut chart filters the scatter plots to that segment only, allowing comparison of how the HP-to-price relationship behaves differently within the \$50K–\$100K tier versus the \$500K+ tier.
- **Year range slider:** A continuous year slider filters the dataset to any production year window, enabling decade-level comparisons or isolation of modern models.

5. Conclusions and Future Work

Key Conclusions

This project set out to answer the question of which performance variables most significantly drive sports car prices. Based on the EDA and correlation analysis conducted across 1,007 vehicles from 38 manufacturers, the following concrete conclusions can be drawn:

- **Horsepower is the strongest individual predictor of price.** With a Pearson correlation of $r = 0.80$ (outlier excluded), horsepower explains a substantial portion of price variation. For most combustion-engine sports cars, every additional 100 HP is consistently associated with a meaningful price premium.
- **A composite performance score outperforms any single metric.** The engineered Final Performance Score (combining normalised HP, torque, and inverted 0-60 time) achieved $r = 0.60$ across the full dataset. This suggests that buyers respond to overall performance capability rather than any single specification.
- **Price distribution is right-skewed and hypercar-driven.** The median price (~\$140K) is a far more representative measure of the sports car market than the mean (\$383K). The extreme tail is driven by fewer than 70 vehicles from Bugatti, Koenigsegg, Pagani, and Rimac, which may warrant separate analysis.
- **Brand identity commands a premium beyond measurable performance.** Pagani averages \$2.79M with 758 HP, while a comparable performance score at a different manufacturer might price at under \$500K. Brand prestige represents a significant pricing variable that is not captured by performance metrics alone.
- **Electric vehicles require separate treatment.** EVs break the engine-size-to-horsepower relationship that underpins much of the dataset's predictive structure. Any future regression model should include an EV/ICE binary control variable.
- **Engine size is a weak standalone predictor ($r = 0.40$).** Contrary to the initial hypothesis, engine displacement is not a reliable price predictor, particularly once electric vehicles are included in the dataset.

Future Work

Several natural extensions of this analysis would increase its depth and practical utility:

- **Linear and log-linear regression modelling:** Building a formal regression model with Price (or log Price) as the dependent variable and HP, Torque, Engine Size, and 0-60 as predictors would quantify the marginal price contribution of each performance unit and allow price prediction for new models.
- **ICE vs. EV segmentation:** Splitting the dataset into combustion and electric subsets and building separate models for each would significantly improve predictive accuracy and produce more interpretable coefficients.
- **Brand fixed effects:** Including brand as a categorical dummy variable in a regression model would allow separation of the brand premium effect from the performance effect, answering how much Bugatti charges above what its performance alone would predict.

- **Real-world data sourcing:** Replacing or supplementing the LLM-generated dataset with actual transaction data from sources such as AutoTrader, Cars.com, or Hagerty would make the findings genuinely prescriptive rather than illustrative.
- **Time-series analysis:** With a longer and more consistent dataset, a time-series model could track how the price-to-performance ratio has evolved decade by decade — investigating whether modern cars deliver more performance per dollar than older generations.
- **Dashboard expansion:** Adding a regression prediction input panel to the Tableau dashboard where a user enters HP, torque, and 0-60 time and receives a predicted price range would transform the dashboard from a descriptive tool into a decision-support tool.

6. References and Acknowledgements

Datasets

- **Primary dataset — Sports Car Prices Dataset:** rkiattisak. (2023). Sports Car Prices Dataset. Kaggle. <https://www.kaggle.com/datasets/rkiattisak/sports-car-prices-dataset>
- **Validation dataset 1 — Car Price Analysis Dataset:** ayeshasiddiqa123. (2023). Cars Pre Dataset. Kaggle. <https://www.kaggle.com/datasets/ayeshasiddiqa123/cars-pre/data>
- **Validation dataset 2 — Vehicle Price Prediction Dataset:** metawave. (2023). Vehicle Price Prediction Dataset. Baselight/Kaggle. https://baselight.app/u/kaggle/dataset/metawave_vehicle_price_prediction

Tools and Software

- **Microsoft Excel:** Used for data profiling, cleaning, pivot table analysis, feature engineering, histogram generation, and correlation calculations using the Data Analysis Toolpak.
- **Tableau Public:** Used for the interactive dashboard. Published at: <https://public.tableau.com/app/profile/navjot.singh1075/viz/2SportsCarPriceAnalysis/Dashboard1>

Analytical Methods Referenced

- **Pearson correlation coefficient:** Used to measure linear relationships between numeric variables. Calculated using Excel's CORREL() function and the Data Analysis Toolpak.
- **Min-max normalisation:** Applied to horsepower, torque, and 0-60 MPH time when constructing the composite performance score.
- **Exploratory Data Analysis framework:** Four-step EDA approach: (1) column selection, (2) individual column exploration, (3) two-dimensional analysis, (4) correlation analysis.